

Chloe Barker

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TECHNICAL SKILLS

Domains: Biological Data, Healthcare Data, Demographic & Public Health Data (CDC, CMS, EPA, NIH)

Data Engineering: Database Design, Normalization, Data Modeling, Query Optimization, Indexing, ETL Pipeline Automation, Advanced T-SQL (Stored Procedures, Views, CTEs, Window Functions, Triggers, Partitioning)

Data Platforms: Databricks, Snowflake, MySQL, MS SQL Server, AWS (S3, EC2), Azure ML

BI & Visualization: Tableau (Dashboards, LOD Expressions, Calculated Fields), Power BI, R Shiny (Interactive Apps), ggplot2, matplotlib, seaborn, HTML/CSS

Data Science & Machine Learning: Python (pandas, NumPy, scikit-learn), R (tidyverse, caret, ggplot2, olsrr), SAS, Classification (KNN, LDA, QDA, Logistic Regression, Random Forest), Regression (Linear, Logistic, LASSO), Feature Engineering, Time Series, Model Selection (AIC/BIC), Cross-Validation, Predictive Modeling

Statistics: Hypothesis Testing, ANOVA, A/B Testing, Experimental Design, Multivariate Statistics

Tools: Git/GitHub, Jupyter Lab/Notebooks, VS Code, MS SQL Server Management Studio, MS Office Suite (Excel, PowerPoint, Word, Visio), Google Suite

Research & Communication: Reproducible Research, Scientific Writing, Experimental Design, Survey Design, Protocol Development, Scientific Communication, Executive Storytelling, Data-Driven Presentations

WORK EXPERIENCE

Applied Analytics Group, LLC | Remote

Jan 2025 - Present

Data Scientist

- Build predictive models and analytics pipelines using Python, R, and SQL to generate client-ready insights.
- Develop dashboards and automated workflows (Tableau, Power BI, Databricks) to support data-driven decisions.

Outlier AI | Remote

Aug 2024 - Jan 2025

AI Math Evaluator

- Evaluated AI mathematical reasoning across algebra, calculus, and statistics to improve model accuracy.
- Designed structured test cases and error analyses to enhance reliability of reasoning-based AI systems.

RUSH University, Department of Microbial Pathogens and Immunity | Chicago, IL

Jun 2023 - Aug 2023

Research Assistant

- Analyzed microbiome and immune-response datasets using Excel and R to study alcohol and circadian effects.
- Cleaned, visualized, and documented experimental data to support reproducible biomedical research.

The University of Texas at Austin, Nutrigenomics Biological Sciences Lab | Austin, TX

Aug 2021 - Dec 2021

Undergraduate Researcher

- Investigated effects of artificial sweeteners on bacterial diversity using microbiome data and statistical methods.
- Conducted trend analysis and visualizations in Excel and R to communicate findings to research teams.

RUSH University, Department of Microbial Pathogens and Immunity | Chicago, IL

Jul 2021 - Aug 2021

Research Assistant

- Analyzed longitudinal datasets on HIV medication impacts using descriptive statistics and data cleaning.
- Built visual summaries and maintained precise documentation supporting ongoing clinical research.

EDUCATION

Southern Methodist University | Dallas, TX

Jan 2025 - Present

M.S. in Data Science (Expected August 2026)

Dallas College | Dallas, TX

Aug 2024 - Dec 2024

- Intro to Python Programming, Calculus I, Statistical Methods

The University of Texas at Austin | Austin, TX

Aug 2020 - May 2024

B.S.A in Biology

Minor in Health Communication | Pre-Health Professions Certificate | GPA: 3.77

- Elements of Data Science, Biostatistics, Management Information Systems, Accounting, Health Communication, Physics I & II, Organic Chemistry I & II

LEADERSHIP & COMMUNITY

Zeta Tau Alpha | *Philanthropy Committee, STAR Team, Member*

Phi Gamma Nu Business Fraternity: Epsilon Beta | *Member*

INTERESTS

Epidemiology, Healthcare Analytics, Running, Outdoor Sports, Psychology, Cooking, Strategy Games, Stand-Up Comedy